

KUMAR CHARAN M

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years of experience designing, developing, and scaling microservices-based backend systems and AWS cloud applications using Java and Python. Proficient in Java 21, Spring Boot, and AWS serverless architecture (Lambda, DynamoDB, API Gateway, SQS, SNS, SES, ECS). Experienced in PCI DSS 4.0 compliance, JWT-based security, event-driven architecture, REST API design, and CI/CD pipelines. Skilled in applying OOP principles and design patterns to build maintainable, high-performance systems. Upskilling in AI/ML development including LLM agents, RAG pipelines, and LlamaIndex frameworks.

TECHNICAL SKILLS

- **Languages:** Java 21 / Jakarta EE, Python 3, SQL
- **Frameworks & Libraries:** Spring Boot, Spring Security, Spring MVC, JPA, Hibernate, AspectJ, Maven, JWT, ConfigCat, FusionAuth, Express.js
- **AWS Cloud:** Lambda, API Gateway, DynamoDB, SQS, SNS, SES, ECS, EC2, S3, IAM, CloudFront, Cognito, CloudWatch, ALB, NLB, VPC, AWS SAM
- **Databases:** PostgreSQL, MySQL, Oracle DB, DynamoDB (NoSQL)
- **API & Integration:** REST APIs, GraphQL, SOAP, JSON, HTTP/HTTPS, Swagger / OpenAPI
- **AI & ML Tools:** LlamaIndex, RAG Pipelines, LLM Agents, Prompt Engineering, Google Gemini, GitHub Copilot, Cursor
- **OOP & Design Patterns:** Object-Oriented Programming, Factory, Singleton, Strategy, Builder, Router, Event-Driven Architecture
- **Testing & Logging:** JUnit 5, Mockito, SLF4J, Log4j, Python logging
- **DevOps & Tools:** Git, Docker, Azure DevOps CI/CD, Postman, IntelliJ IDEA, VS Code, SoapUI, GitHub Actions, Rancher Desktop, WSL, Ubuntu
- **Methodologies:** Agile (Scrum), Microservices Architecture, Cloud-Native Development, Serverless Architecture

WORK EXPERIENCE

Senior Software Engineer

January 2026 – Present

Infomerica Inc India Pvt. Ltd · US-Based Energy Client

- Owned end-to-end architecture, development, and delivery of Java/Spring Boot microservices and AWS serverless solutions (Lambda, API Gateway, DynamoDB, SQS, SNS, SES, ECS) provisioned via AWS SAM.
- Reduced cold start latency by 50%+ across shared platform Lambda functions using Provisioned Concurrency, Event-Driven Architecture, Router Design Pattern, and API Gateway stage caching for high-traffic GET endpoints.
- Improved dashboard initial API load time from 6s to 2s (67% improvement) by optimizing the critical first API call and developing an Event-Driven Notification API triggered on user login.
- Implemented JWT-based Custom Authorizer Lambdas and .p12 certificate-based secure communication channels to enforce PCI DSS 4.0 compliance across banking and energy client applications.
- Led technical architecture decisions, conducted peer code reviews, and mentored junior developers on AWS and Spring Boot best practices within Agile sprint cycles.

Key Projects:

PCI 4.0 Compliance — Tokenization & Authorization Platform

Tech Stack: Python 3, boto3, Node.js, Next.js, Express.js, AWS Lambda, HTTP API Gateway v2.0, SAP Systems

- Developed a Python-based AWS application integrating a third-party tokenization tool to encode credit card and bank account numbers, generate secure tokens, and issue JWT tokens for downstream transaction authorization.
- Built a Node.js Lambda Authorizer for HTTP API Gateway v2.0 to handle PCI DSS 4.0-compliant authorization for all inbound requests.
- Engineered a secure Express.js proxy between SAP systems and a third-party provider using digitally signed .p12 certificate-based mutual authentication.

Energy Provider OAM Cloud Migration

Tech Stack: Java 21, Spring Boot, AWS Lambda, DynamoDB, API Gateway, CloudWatch, Oracle DB, Maven, AWS SAM

- Built a brand-specific cloud application from scratch, developing and deploying all AWS services via AWS SAM.
- Architected a JWT-based Custom Authorizer Lambda on AWS API Gateway — validating access tokens and JWT claims against DynamoDB to minimize unauthorized access.

Common OAM Cloud Platform (Multi-Brand Shared Platform)

Tech Stack: Java 21, Spring Boot, AWS Lambda, DynamoDB, SQS, SNS, SES, ECS, EC2, CloudWatch, Oracle DB, Maven, AWS SAM

- Built a shared private internal cloud platform from scratch serving all client brands, covering Lambda, DynamoDB, SQS, SNS, SES, ECS, and EC2, deployed via AWS SAM and Azure DevOps CI/CD pipelines.
- Designed asynchronous event-driven workflows using SQS, integrated DynamoDB-backed feature flags via ConfigCat, and connected third-party APIs for real-time email validation.

Software Engineer

January 2024 – December 2025

Infomerica Inc India Pvt. Ltd · US-Based Energy Client

- Designed and developed cloud-native AWS Lambda functions and Spring Boot REST APIs (JSON/HTTP) applying OOP principles and design patterns including Factory, Singleton, and Strategy.
- Built a shared internal cloud platform from scratch serving multiple client brands, deployed via AWS SAM and Azure DevOps CI/CD pipelines.
- Implemented asynchronous event-driven workflows using SQS, SES, and SNS; integrated Experian API for real-time email validation and ConfigCat for DynamoDB-backed feature flag management.
- Collaborated across Agile cross-functional teams — frontend, infrastructure, QA, and product — to deliver end-to-end features across multiple client brands.

Key Projects:

Energy Client Texas Application — Microservices Migration

Tech Stack: Spring Boot, Spring MVC, AWS Lambda, ECS, EC2, ALB, Mockito, Git, Maven

- Migrated microservices to AWS ECS on EC2 instances organized under distinct target groups managed through an Application Load Balancer (ALB).
- Re-architected a monolithic Tomcat-based Spring application into a containerized Spring Boot microservice. Added Mockito test coverage for all Lambda integrations.

Energy Client Canada Applications — Modernization

Tech Stack: Spring Boot, Spring MVC, Oracle DB, Maven, REST APIs, jQuery

- Modernized two legacy Spring applications into Spring Boot — rewrote Oracle DB queries, developed new modules, redesigned UI components with jQuery, and wrote regression test cases.

Associate Software Engineer

July 2022 – December 2023

Infomerica Inc India Pvt. Ltd

- Migrated legacy monolithic Tomcat-based Spring applications to containerized Spring Boot microservices deployed on AWS ECS with Application Load Balancer (ALB) routing.
- Delivered CMS integration across 3 enterprise applications (1 B2C, 2 B2B) — replaced legacy XML-based APIs with AEM CMS GraphQL queries, standardizing responses from XML to JSON with zero regression.
- Worked with Oracle DB, JPA, and Hibernate for data persistence, query development, and schema management across multiple applications.

Key Projects:

CMS Migration for Enterprise Client Systems

Tech Stack: Java 8 & 11, Tomcat 9, Maven, AEM CMS, GraphQL, SOAP, REST APIs

- Delivered a CMS integration layer across 3 enterprise applications (1 B2C, 2 B2B) — replaced legacy XML-based CMS APIs with AEM CMS GraphQL queries.
- Standardized all API response formats from XML to JSON with zero regression across all applications.

Software Trainee

January 2022 – June 2022

Infomerica Inc India Pvt. Ltd

- Built and consumed REST APIs using Spring Boot within a microservices architecture; implemented Spring Security with JWT-based authentication and role-based access control (RBAC).

- Applied JPA and Hibernate for entity mappings and database persistence; implemented AOP for cross-cutting concerns; wrote unit tests using JUnit 5 and Mockito.

PERSONAL PROJECTS

Smart Note — AI-Powered Notes Analyser & Converter

April 2026 – Present

Tech Stack: Python, LlamaIndex, RAG, Vector Indexing, LLM Agents, Prompt Engineering

- Building a RAG pipeline in LlamaIndex covering document ingestion, vector indexing, and grounded Q&A with source attribution.
- Implemented function-calling to enable an LLM agent to autonomously summarize, extract key points, improve text, and translate across parsed document nodes using structured prompt templates.

GitHub PR Review Agent

March 2026 – April 2026

Link: github.com/KumarCharan-00/PR-Review-Agent

Tech Stack: Python 3.10, LlamaIndex AgentWorkflow, Google Gemini, PyGithub, GitHub Actions

- Built a multi-agent AI system that automatically reviews Pull Requests — fetching PR context, drafting a review, self-validating it, and posting it as a formal GitHub review comment.
- Architected a 3-agent pipeline (ContextAgent, CommentorAgent, ReviewAndPostingAgent) using LlamaIndex AgentWorkflow with a shared state store. Supports both local testing and automated CI/CD via GitHub Actions.

ISKM Proddatur Website

November 2025 – March 2026

Link: dev.iskmpdtr-int.cloud

Tech Stack: Spring Boot, PostgreSQL, Docker, Vanilla JS, React, Dokploy

- Built and deployed a full-stack web application: Spring Boot + PostgreSQL backend API for content management and a React frontend for public display with a Vanilla JS admin interface, all containerized with Docker and deployed on Dokploy.

EDUCATION

Bachelor of Technology (B.Tech)

June 2019 – April 2023

Anurag Group of Institutions, Ghatkesar, Hyderabad